Problem Chosen	2024 MCM/ICM Summary Sheet	Team Control Number
B		2424101

Into the Depths Unseen: Innovating Submersible Tracking

Abstract

Maritime Cruises Mini-Submarines (MCMS), a Greek company dedicated to the construction of manned submersibles, proposes to open a business of submersible undersea hull wreck tours in the Ionian Sea, which requires the establishment of a safety procedure to cope with the difficulties of lost submersibles or mechanical failures. In response, we developed 3 models: a submersible position prediction model, a search equipment planning model, and a search pattern optimization model. Also, we consider scalable extensions for different sea areas and multi-submersible complexities.

Submarine Position Prediction Model (Model 1): Firstly, considering factors like ocean currents and seawater density, the hydrodynamic model of the submersible is established from kinematics and dynamics. Then, based on known current data of 100 positions, 10,000 current flow rate scenarios are simulated using the Bootstrap method, estimating mean and standard deviation intervals. Finally, relevant parameters are substituted into the hydrodynamic model's differential equations to determine the sub's free motion mode with ocean currents, finding its eventual stable depth value.

Search Pattern Optimization Model (Model 2): Our group collects various types of search equipment, along with their attributes such as rescue performance and usage cost, aiming to rationalize the combination of various equipment for the main vessel. Besides, we consider 72 combination schemes and use the entropy weighting method to determine the weights of different attributes for these combinations. Finally, our group employs the TOPSIS method to comprehensively evaluate and rank these 72 schemes, then we provide the top five optimal combination schemes.

Search Pattern Optimization Model (Model 3): First, based on model 1, the Monte Carlo algorithm is used to simulate the possible position of the submersible one hour after it is lost. Then the convex packet algorithm is used to determine the search area which is gridded; then the genetic algorithm is used to obtain the optimal solution of the search path by using the time required for a grid search of the SAR vessel as the fitness function, and the search path is obtained by iterating for 100 times. Finally, based on Bayesian search theory, combined with Koopman's probability function in search theory to achieve the preferred effect of real-time updating search path.

In addition, our group discusses the changes needed to apply the above models to the Caribbean Sea, taking into account the special local natural geographic conditions of the Caribbean Sea. For the problem of simultaneous operation of multiple submersibles, this group gives an initial deployment plan by considering the safety and experience of underwater exploration and discusses the joint search and rescue model under the matching mechanism of four submersibles as a unit in case of emergency.

Finally, the robustness and sensitivity analyses of the model are examined: when the initial distribution of the current velocity magnitude is generated randomly, the final converged distributions of the model do not differ much. For the variation of drag coefficients C_d, the displacement of the submersible tends to change gently.

Keywords: Hydrodynamic model, TOPSIS, Genetic Algorithm, Bayesian search theory

Content

1 Introduction

1.1 Problem background

A Greek company called Maritime Cruising Mini Submarines (MCMS) manufactures submersibles that offer tourists the possibility of exploring the remains of shipwrecks on the Ionian seabed and it needs to build a reliable safety program that will be approved by regulatory agencies.

In the event that the submersible is involved in an accident and loses communication with the main vessel, the company hopes to build a model that will effectively predict the position of the submersible over time, facilitating subsequent rescue operations.

It is worth noting that a malfunctioning submersible may find itself on the seafloor or at some point of neutral buoyancy underwater, and at the same time its position will be affected by ocean currents, varying seawater densities in various parts of the oceans, and seafloor topography, which is markedly different from search and rescue conducted on land or at the surface of the sea.

1.2 Restatement of the problem

Considering the background information and constraints identified in the problem statement, we need to address the following questions:

- What are the components of uncertainty associated with this prediction?
- What equipment does the submersible need to minimize this uncertainty?
- What additional equipment might a rescue vessel need to carry to assist if required?
- How to build a function that the probability of finding a diver is presented as a function of time and cumulative search results?
- How can your model be extended to other destinations, such as the Caribbean?
- How will the model change to when multiple submersibles move in the same vicinity?

1.3 Our work

Here is our thought process for the whole questions:

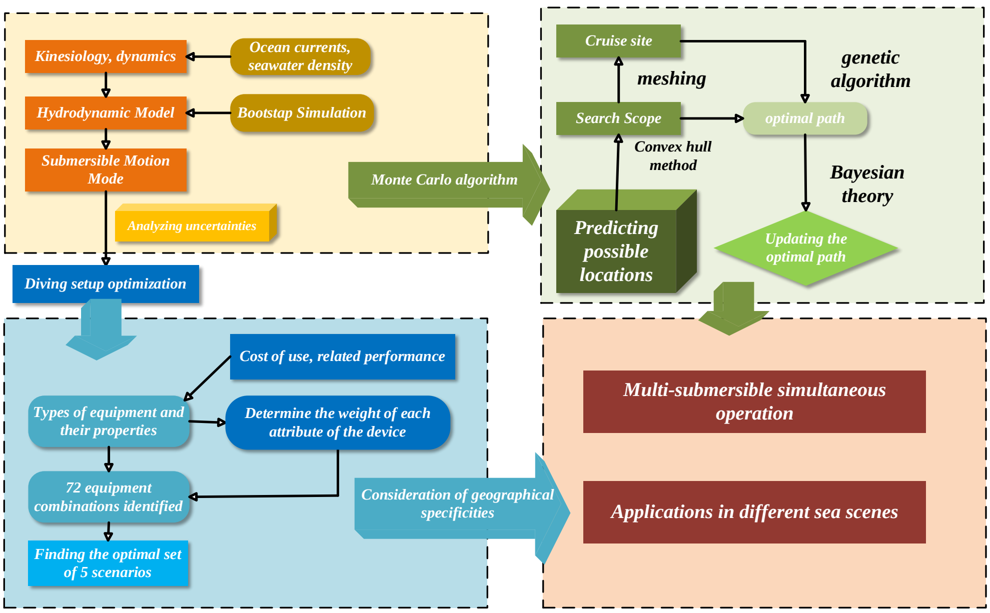


Figure 1 Work Thinking Flowchart

2 Assumptions and Justifications

- **Assumption 1: The density of seawater and the speed of ocean currents do not change over time.**

Justification: These factors change relatively little on short time scales or in particular regions.

- **Assumption 2:The submersible is a standard cylinder (length , bottom diameter).**

Justification: Treating the submersible as a cylinder makes it easier to calculate the effects of forces such as buoyancy and drag on it.

- **Assumption 3:**The position of the submersible is independent of its attitude.

Justification: We are primarily concerned with the approximate position and path of the submersible, not its precise orientation or attitude dynamics in the water.

- **Assumption 4:** The velocity of the ocean current does not contain a component in the vertical direction.

Justification: Such a simplifying assumption helps to focus on the motion and distribution of ocean currents in the horizontal plane and, in many cases, horizontal motion has a much greater impact on the ocean dynamics system than vertical motion.

- **Assumption 5:**Assume that the submersible loses propulsion completely after a mechanical failure.

Justification: We assume that the submersible moves freely with the ocean currents to facilitate the construction of the dynamic equations of the submersible.

- **Assumption 6:**It is assumed that the study data are accurate.

Justification: We assume that there is no significant deviation in the data related to ocean currents, temperature, and salinity, and therefore we are able to build a more reasonable quantitative model on this basis.

3 Notations

3.1 Important Symbols

The key mathematical notations used in this paper are listed in Table 1.

Table 1 Important symbols

Symbol	Description	Unit
$\boldsymbol{S}$	Rate of an ocean current	m/s
ρ	Water density	Kg/m^3
$\boldsymbol{V}$	Submersible speed	m/s
$L(x,y,z)$	Submersible location	
θ	Angle of the current speed with the axis	rad
φ	Submersible speed and axis angle	rad
C_d	Drag coefficient	
$P(D)$	Probability of finding the submersible	
P	Pressure	Pa
S	Salinity	g/kg
T	Temp	K

3.2 Digital

The literature and data reviewed in this paper are listed in Table 2

Table 2 Data sources

Database Name	Database Website	Data Type
CNN	httpts:www.cnki.net	paper
Google Scholar	https://scholar.google.com/	paper
PO.DAAC	https://podaac.jpl.nasa.gov/cloud-datasets/dataaccess	geographic data
EU WEkEO	https://www.wekeo.eu/data	geographic data
GEBCO	https://www.gebco.net/data_and_products/gridded_bathymetry_data/	geographic data
Ocean OPS	https://www.ocean-ops.org/board	geographic data

4 Model Preparation

4.1 Block processing of seabed topography in the target area

The Ionian Sea, as a part of the Mediterranean Sea, has a complex seafloor topography. In the modeling process of the thesis, the topography of the Ionian seabed has a more important influence. In order to effectively incorporate the seabed topography into our model building process, and to reduce the interference of the complex and variable topography on the model building to a certain extent, according to the color depth of the seabed topography map of the sea area, the seabed topography is roughly divided into three parts: A , B , C .

The A section is nearer to land, at lower depths, but the terrain is more complex; the B section is undulating with many gullies; and the C section is farther from land, at deeper depths, but the terrain is less varied than that of the A and B sections.

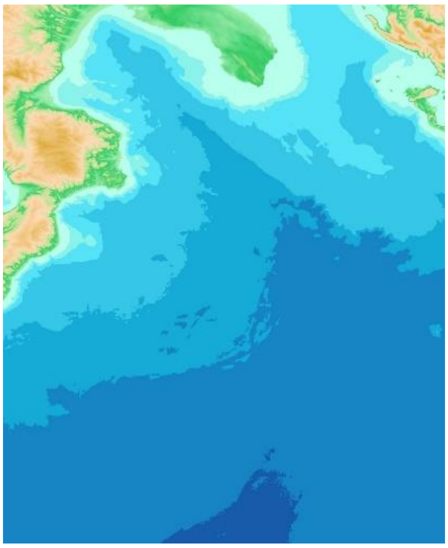


Figure 2 (a) Topographic map of the Ionian seabed

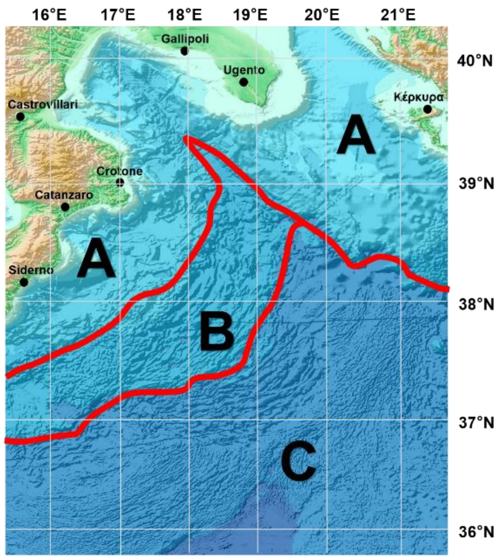


Figure 2 (b) Topographic map

4.2 Establishment of the coordinate system

In order to obtain the real distance relationship in the map, we approximate the observed map area as a quadrilateral. After solving for the length of this quadrilateral using the geodesic equation (GRS80 sphere), the correspondence between spherical and planar coordinates is fitted by a projection transformation, with due east as the direction of the x-axis, due north as the direction of the y-axis, and the z-axis perpendicular to the plane determined by the x-axis and y-axis. In this way, the Euclidean distance between points approximates the geodesic distance on the sphere. It is worth noting that in this process, we only consider three degrees of freedom [1] and do not include the attitude of the submarine in the consideration of model building, which will maximize the practicality and effectiveness of the model.

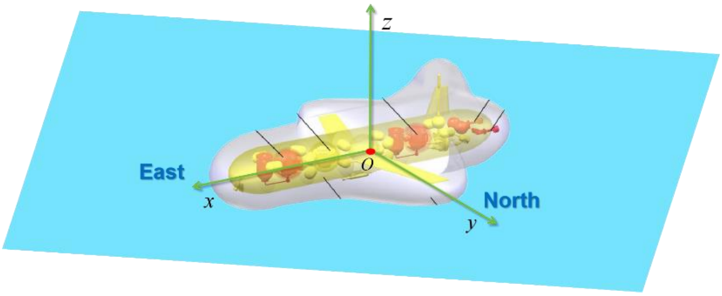


Figure 3 HOV coordinate system

5 Model I: Submersible Position Prediction Model

The position of a submarine needs to be considered from two perspectives: one is based on kinematics to explore the relationship between the change in position of the submarine and its velocity, and the other is based on dynamics to explore the relationship between the force on the submarine and its acceleration. By combining these two aspects, a dynamic prediction model of the position of the submarine can be obtained.[1]

Given that the focus of the study is on the position change of the submersible rather than its attitude (orientation) change, this group will only consider the three translational degrees of freedom of the submersible (longitudinal, lateral, and vertical movement) and ignore the rotational degrees of freedom, i.e., the acceleration and angular acceleration of the submersible will not be considered.

The model will model the dynamics from the longitudinal movement, lateral movement, and vertical movement of the submersible respectively.

5.1 Kinematic modeling of submersibles

To discuss the problem, our group gives the main parameters and symbols of the submersible according to the common naming rules. Let the velocity of the center of gravity of the submersible relative to the earth be V. In the coordinate system, the components of the velocity along the three coordinate axes are the longitudinal velocity X, transverse velocity Y, and vertical velocity Z, the components of the force along the three coordinate axes are the longitudinal force, transverse force, and vertical force; and the states of motion along the three coordinate axes are the inward and outward motion, the sideways motion, and the submerged motion, respectively.

Table 3 Common naming rules

	$O-xyz$ system		
	x axis	y axis	z axis
Speed	u	v	ω

Force	X	Y	Z
State	Advance and Back	Lateral Movement	Snorkel

From the kinematic point of view, the position update condition can be calculated after determining the velocity of the submersible's motion. We can get the expression for the position of the submersible with respect to the velocity as $(dx, dy, dz) = \boldsymbol{V}dt$. We consider the initial velocity of the submersible $V\left(u, v, \omega\right) = 0$, and the initial coordinates $L\left(x_0, y_0, z_0\right) = 0$.

5.2 Dynamic modeling of submersibles

5.2.1 The role of ocean currents on submersibles

If the velocity of the current is known, the force of the current on the submersible F [1] can be expressed as:

$$F = \frac{1}{2} C_d \rho A \nabla V^2 \tag{1}$$

Where A is the cross-sectional area of the submersible directly in the direction of the fluid, C_d is the drag coefficient, ρ is the density of the seawater, and ∇V is the velocity of the current relative to the hull.

Assuming that the direction of the velocity of the current makes an angle with the axis of θ, the velocity of the current can be expressed as:

$$\boldsymbol{S} = (S\cos\theta, S\sin\theta) \tag{2}$$

In the constructed submersible dynamics model, θ is the random number on $[-\frac{\pi}{6}, \frac{\pi}{6}]$, transformed into the corresponding angular system, i.e., $[-30^\circ, 30^\circ]$.60 different sets of θ are taken with 1° as the step size to prepare for the subsequent simulation of the corresponding trajectory positions of different submersibles respectively.

At depths of about 2000 meters, the usual current speed is 5-10 cm/s. The current speed decays exponentially with depth.

$$S_h = S_0 e^{-\frac{\pi}{D_a}h} \tag{3}$$

where h is the water depth, S_0 is the current speed at the surface of the current, D_a is the friction depth, and S_h is the current speed when the depth is h.

Since the speed of the submersible in the horizontal plane $\left(u\left(t\right), v\left(t\right)\right)$, the direction of the speed of the submersible makes an angle with the x-axis:

$$\varphi = \arctan(\frac{v(t)}{u(t)}) \tag{4}$$

So:

$$\nabla V = S_h - \sqrt{v(t)^2 + u(t)^2}\cos(\theta - \varphi) \tag{5}$$

In calculating the cross-sectional area A directly opposite the flow direction, since the current acts partly on the sides of the cylinder and partly on the bottom of the cylinder, we approximate that there is:

$$A = ld\sin(\theta - \varphi) + \pi(\frac{d}{2})^2 \cos(\theta - \varphi) \tag{6}$$

The drag coefficient C_d is a dimensionless measure of the resistance of an object to flow, which is affected by the shape of the object, the roughness of the surface, and the nature of the current.

In summary, the force of the current on the submersible can be expressed as:

$$\begin{cases} F_x = \frac{1}{2} C_d \rho A \nabla V^2 \cos\theta \\ F_y = \frac{1}{2} C_d \rho A \nabla V^2 \sin\theta \end{cases} \tag{7}$$

5.2.2 The role of seawater density on submersibles

For the sake of simplicity, the phenomenon of the "sea cliff" (a sudden drop in the density of seawater as the depth decreases) is not taken into account. We use the equation of state of seawater, which is an equation that describes the relationship between the density of seawater, ρ, and the temperature, T, the salinity, S (i.e., the amount of salt in the seawater), and the pressure, P (the depth).

The density of seawater ρ can be expressed as a function of temperature T (℃), salinity S (absolute salinity unit: pus) and pressure P (pa):

$$\rho = f(T, S, P) \tag{8}$$

This question uses an empirical formula to approximate the density of seawater:

$$\rho(T, S, P) = \rho_0 + a(T - T_0) + b(S - S_0) + c(P - P_0) \tag{9}$$

The values ρ_0 、 T_0 、 P_0 are the values of reference density, temperature, salinity and pressure, respectively (generally the relevant data at the sea surface), and a , b and c are the coefficients determined from the experimental data.

For the general trend of seawater temperature with depth h can be approximated using an exponentially decreasing model as:

$$T = T_0 - k_1 e^{-bh} \tag{10}$$

Where T_0 is the temperature of seawater, k_1 and b are constant parameters.

The variation function of seawater salinity can be approximated as:

$$S = S_0 - k_2 h \tag{11}$$

where the coefficient k_2 is a coefficient of variation.

It is important to note that when considering deep sea applications, the pressure P is estimated from the actual depth of the dive h and the average density of the seawater ρ_0:

$$P - P_0 = \rho_0 g h \tag{12}$$

Substitutions can be made:

$$\rho(h)=\rho_0+\lambda e^{-bh}+\mu h+\eta h \tag{13}$$

Among them:

$$\lambda=-k_1a \qquad \mu=-k_2b \qquad \eta=\rho_0gc \tag{14}$$

By the general buoyancy formula:

$$F=\rho Vg \tag{15}$$

The buoyant force on the submersible in the Z -axis direction F is:

$$F=(\rho_0+\lambda e^{-bh}+\mu h+\eta h)g\pi(\frac{d}{2})^2l \tag{16}$$

5.2.3 Viscous hydrodynamics and inertial hydrodynamics

The viscous hydrodynamics of a submersible under the sea is mainly caused by the viscosity of the seawater. Deep-sea submersibles can generally be considered to operate in an infinite fluid, so the viscous and inertial hydrodynamic forces depend only on the velocity and acceleration of the motion.

The rate of change of the motion of the submersible is not considered, and the viscous hydrodynamic force it is subjected to during its voyage is only related to the current state of motion, and is not related to the historical state of motion. The expression for the viscous hydrodynamic force on the submersible is considered here[2] :

$$\begin{cases} f_x=\dfrac{\rho l^2}{2}[X_{uu}u^2+X_{vv}v^2+X_{\omega\omega}\omega^2] \\ f_y=\dfrac{\rho l^2}{2}Y_{v\omega}v\omega \\ f_z=\dfrac{\rho l^2}{2}Z_{vv}vv \end{cases} \tag{17}$$

When a submersible moves at a variable speed in seawater, such as acceleration, the fluid plasmas in the flow field around the object transfer energy and also dissipate some of that energy for use in accelerating its own acceleration. The resulting inertial hydrodynamic force is proportional to the magnitude of the velocity of the submersible and in a direction opposite to that of the velocity, in the following form:

$$\begin{cases} F_x=-\lambda_{11}u \\ F_y=-\lambda_{11}v \\ F_z=-\lambda_{11}\omega \end{cases} \tag{18}$$

λ_{11} It is called the added mass, also known as the scale factor.

5.2.4 Dynamic equations for submersibles

Combining the action of ocean currents on the submersible, the force of seawater density on the submersible, and the viscous and inertial hydrodynamics of the submersible, the dynamic equations of the submersible are:

$$\begin{cases} X = m\cdot\dfrac{d^2x}{dt^2} = \dfrac{1}{2}C_d\rho A\nabla V^2\cos\theta + \dfrac{\rho l^2}{2}[X_{uu}u^2 + X_{vv}v^2 + X_{\omega\omega}\omega^2] - \lambda_{11}u \\ Y = m\cdot\dfrac{d^2y}{dt^2} = \dfrac{1}{2}C_d\rho A\nabla V^2\sin\theta + \dfrac{\rho l^2}{2}Y_{v\omega}v\omega - \lambda_{11}v \\ Z = m\cdot\dfrac{d^2z}{dt^2} = (\rho_0 + \lambda e^{-bz} + \mu z)g\pi(\dfrac{d}{2})^2 l + \dfrac{\rho l^2}{2}Z_{vv}vv - \lambda_{11}\omega - mg \end{cases} \tag{19}$$

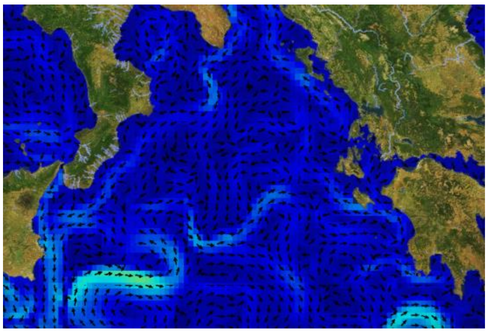
Table 4 Parameter estimation results

Parameters	Value	Unit
D_a	32	m
C_d	0.42	—
λ_{11}	0.432	—
ρ_0	1.025	kg/m^3
λ	0.18	—
μ	0.57	—
η	0.31	—
b	0.83	—

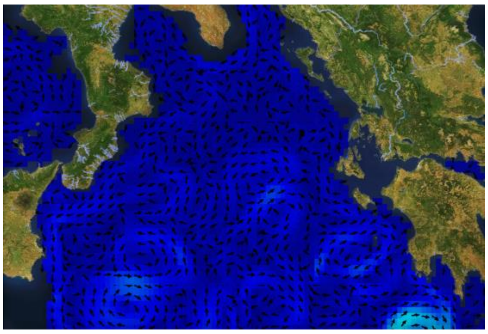
5.3 Modeling results

Let the overall mean value of the current velocity at the sea surface be μ and the overall standard deviation be σ. Since the distribution of the current velocity may be affected by a variety of factors, including wind, water temperature, salinity, seafloor topography, and the rotation of the Earth, etc., it is difficult to represent the actual distribution of the current velocity in a general mathematical model. Therefore, it is difficult to represent the actual distribution of ocean current velocity by a general mathematical model.

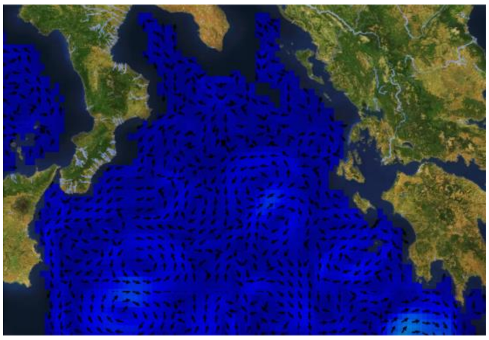
The Bootstrap method estimates the distribution of a statistic (e.g., mean, variance) by taking a large random sample with put-backs from a limited set of raw data. This method does not rely on specific distributional assumptions about the data and is suitable for analyzing complex data sets.



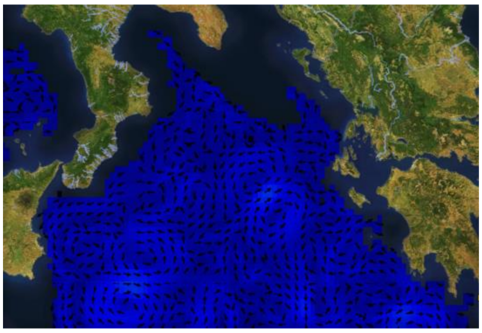
（a）0m



（b）500m



(c)1000m



(d)1500m

Figure 4 Ocean current velocity heatmap

We randomly take the flow size of 100 points from the graph as sample data, and use the put-back sampling method to sample successively and independently from these 100 sample data to obtain 10000 Bootstrap samples of capacity 100. Bootstrap confidence intervals were computed for the mean μ and standard deviation σ simulations, respectively, and the sample mean μ was used as an estimate of the overall mean, and the sample standard deviation s was used as an estimate of the overall standard deviation.

For each Bootstrap sample, the sample mean $\overline{x}_i^*(i=1,2,\cdots 10000)$ is calculated, and the 10,000 $\overline{x}_i^*$ are ranked from smallest to largest, with $\overline{x}_{(500)}^* = 0.028$ being the 500th from the left, and $\overline{x}_{(9500)}^* = 0.041$ being the 9500th from the left. A Bootstrap confidence interval for μ with a confidence level of 0.90 is then obtained as:

$$(\overline{x}_{(500)}^*, \overline{x}_{(9500)}^*) = (0.028, 0.041) \tag{20}$$

Similarly a confidence interval for Bootstrap with a confidence level of 0.90 can be obtained for σ :

$$(s_{(500)}^*, s_{(9500)}^*) = (1.3\times10^{-5}, 2.6\times10^{-5}) \tag{21}$$

Monte Carlo simulations of the magnitude of ocean current speeds are performed using the rand( ) function in Python to generate a large number of random numbers representing the magnitude of the current speeds.

Finally, the data were substituted into the dynamics model of the submersible to obtain the change in position of the submersible within one hour of the wreck.

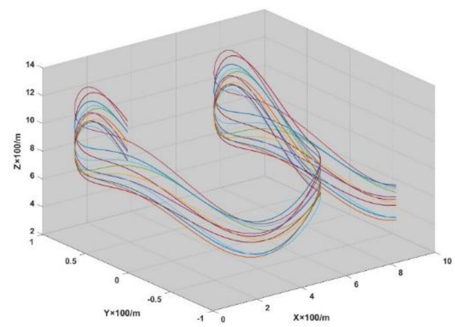


Figure 5 (a) Three-dimensional map of the trajectory

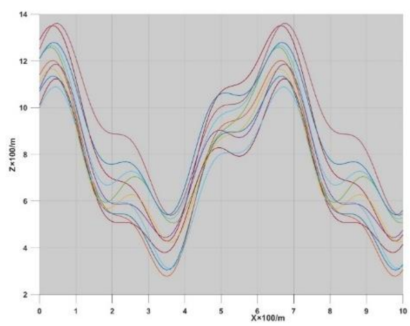


Figure 5 (b) Tracking error

As can be seen from the two figures above, the final depth region of the submersible is smooth and the submersible reaches the stabilization zone.

5.4 Uncertainty analysis

The model does not take into account the specifics of changes in the geographic environment of the seabed; rocks and wreckage on the seabed may obscure the submersible, making search and rescue efforts more difficult in the absence of accurate topographic data. Changes in bottom topography and water layers can affect the detection effect of underwater sonar, and acoustic reflection and refraction phenomena in areas with complex seafloor geomorphology may lead to misjudgments.

Temperature and salinity also differ horizontally, resulting in a water density field that is not only depth-dependent in practice, but varies with spatial location.

The direction of ocean currents is arbitrary, and the above modeling discusses only some of these scenarios and does not provide a comprehensive enough prediction of the likely location area of the submersible.

5.5 Diver precautions

The submersible shall send the precise speed and acceleration of the submersible to the primary vessel at regular intervals;

The current approximate horizontal coordinates of the submersible;

The approximate current depth of the submersible;

Water temperature, pressure, and salinity outside the submersible;

The state of health of the submersible's mechanical systems;

For this purpose, a submersible requires the following equipment: GPS (Global Positioning System), underwater sonar, water temperature sensors, water pressure sensors, engine equipment sensors, and a backup propulsion system.

6 Model II: Submersible Equipment Equipping Model

6.1 Main ship equipped

We have considered the following types of search devices from the function of search:

- Search and Rescue Equipment: ROV,AUV.
- sonar system: ES,SSS, MBS, SE.
- Submersible for search and rescue: UAV,USV.

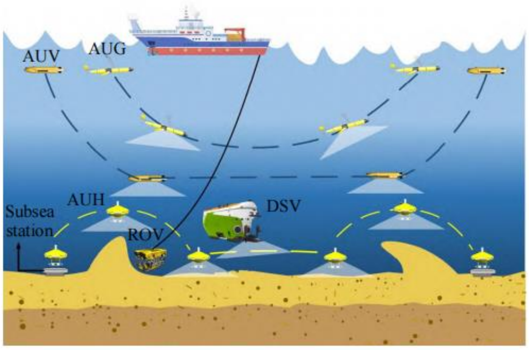


Figure 6 (a) Collaborative operation of AUVs, AUGs and DSVs

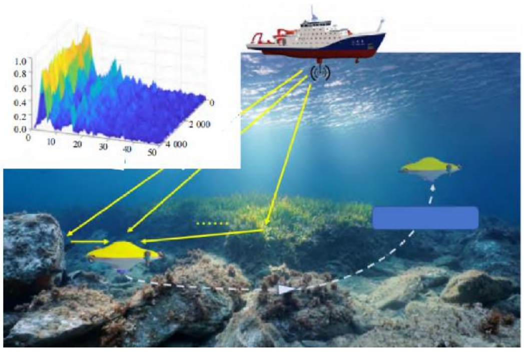


Figure 6 (b) Rescue of main ship and submersible together

6.2 TOPSIS evaluation based on entropy weighting method

We consider that each main ship equipment is equipped with 4×3×3×2=72 combination schemes, and also consider the search and rescue range of the SAR equipment x_1 ,the search accuracy x_2, the search reliability x_3, and the sum of its equipment maintenance cost, equipment preparation cost, and equipment use cost x_4.

The TOPSIS algorithm based on the entropy weight method [2] is used to construct the positive and negative ideal solutions of the evaluation problem, and the optimal solution is selected by calculating the relative closeness of each solution to the comprehension solution and ranking the solutions.

Based on the existing 16 scenarios and the corresponding 4 indicators, the decision matrix $A=(a_{ij})_{72\times 4}$ is obtained.

Using the 0-1 transformation, for the benefit type attribute x_1, x_2, x_3.

Make $b_{ij}=\dfrac{a_{ij}-a_j^{\min}}{a_j^{\max}-a_j^{\min}} \quad (j=1,2,3)$.

For the cost-based attribute x_4, make $b_{i4}=\dfrac{a_4^{\max}-a_{i4}}{a_4^{\max}-a_4^{\min}}$.

Vector normalization is performed for each of the four attributes:

$$\tilde{b}_{ij}=b_{ij}/\sqrt{\sum_{i=1}^{72} b_{ij}^2}\ ,i=1,2,\cdots,72,\ j=1,2,3,4 \tag{22}$$

We solve the weight vector according to the entropy weighting method, currently for 72 the programs to be evaluated, 4 rating indicators, calculate the probability matrix P , where P each element in p_{ij} is calculated by the formula:

$$p_{ij}=\frac{c_{ij}}{\sum_{i=1}^{72} c_{ij}} \tag{23}$$

The information entropy e_j formula for the j ,the indicator is:

$$e_j=-\frac{1}{\ln 72}\sum_{i=1}^{72} p_{ij}\ln(p_{ij})(j=1,2,3,4) \tag{24}$$

Definition of information utility value: d_j =1—e_j ,then the larger the information utility value d_j is, the more information it corresponds to. By normalizing the information utility value d_j, we can get the entropy weight of each indicator ω_j :

$$\omega_j=d_j/\sum_{j=1}^{4} d_j,\ j=1,2,3,4。 \tag{25}$$

Construct the weighted norm matrix C :

$$C = \left(c_{ij}\right)_{72\times 4} \tag{26}$$

$c_{ij} = \omega_j \cdot \tilde{b}_{ij}$， $i = 1, 2, \cdots 72$， $j = 1, 2, 3, 4$。

Then calculate the distances to the positive and negative ideal solutions for each of the 72 solutions:

① Alternative i the distance to the positive ideal solution is:

$$s_i^* = \sqrt{\sum_{j=1}^{4}(c_{ij} - c_j^*)^2}\ ,\ i = 1, 2, \cdots, 72 \tag{27}$$

② Alternative i the distance to the negative ideal solution is:

$$s_i^0 = \sqrt{\sum_{j=1}^{4}(c_{ij} - c_j^0)^2}\ ,\ i = 1, 2, \cdots, 72 \tag{28}$$

Calculate the value of the queuing metric for the i ,the program:

$$f_i^* = s_i^0 / (s_i^0 + s_i^*)\ ,\ i = 1, 2, \cdots, 72 \tag{29}$$

In descending order of f_i^* , the top five options are taken as the available options.

Table 5 Five optimized scenarios

Device number and combinatorial	**equipment cost (\$)**	**Search Scope**	**Search accuracy**	**Search Reliability**
P1+B3+S2+SU1	13900	9.8	48	81
P2+B2+S1+SU2	13800	9.6	45	66
P1+B3+S3+SU2	14700	10	49	64
P2+B1+S2+SU2	13900	8.9	47	60
P1+B2+S3+SU1	13700	8.3	46	77

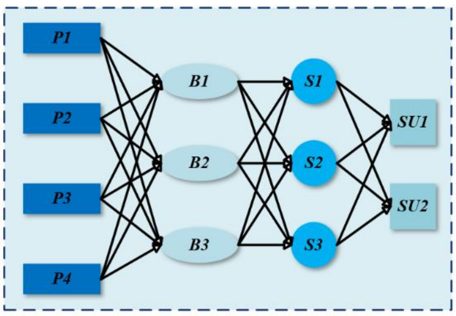


Figure 7 (a)
Equipment mix program

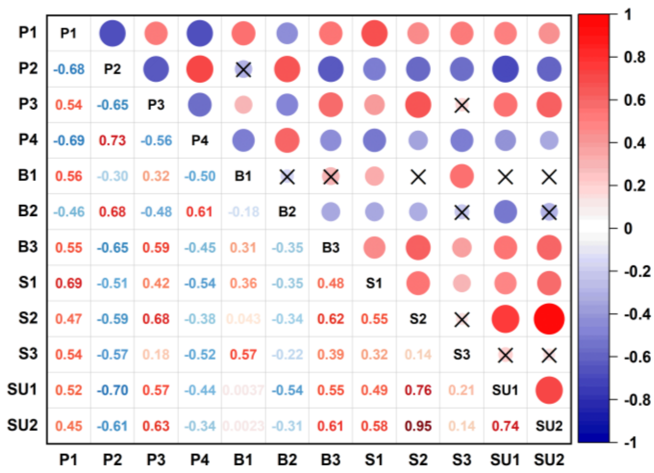


Figure 7 (b) Correlation coefficient graph
(Level of significance: 0.05)

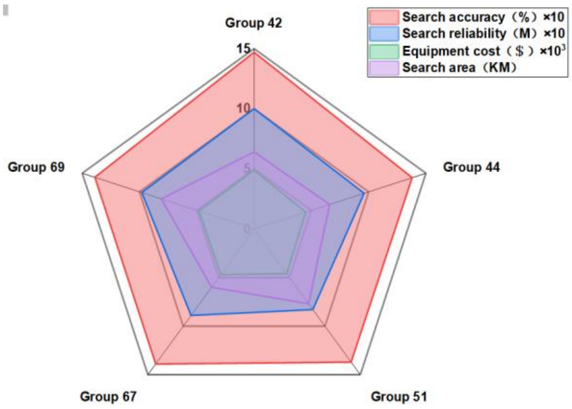


Figure 8 Attribute Radar Chart

The analysis leads us to the final choice of option No. 69.

6.3 Search and rescue vessel equipped

A response to what additional equipment a rescue boat may need to carry to provide assistance is presented here in table form:

Table 6 SAR equipment

Type	Designation	Function	Weight(kg)	Price($)
Life-saving equipment	Life raft	Provide temporary shelter	15	500
	Life buoy	Provides buoyancy and protection	2	50
	Life line	Salvage and towing	1	20
Lighting signal device	Search lights	Bright rescue field of view	3	100
	Flashing lights	Guide search and rescue	1	30
	Flares	Produce light and sound to determine location	0.5	10
Medical equipment and medicines	First-aid case	Treat first aid situations	2	50
	Respiratory apparatus	Deal with dyspnea or asphyxia to ensure the patient's respiratory function	5	200
	Infusion equipment	Provide patients with fluid supplement and drug therapy	3	150
Diving and underwater equipment	Diving suit	Underwater manual rescue	10	500
	Underwater camera	Expand search field and monitor the seabed in real time	2	300
Fire -fighting equipment	Fire pump	Quickly extinguish and control the fire	20	400
	explosion-proof equipment	Prevent the device from generating sparks, arcing, or high temperatures to avoid causing an explosion	15	800

7 Model III: Grid search model based on genetic algorithm

7.1 Initial deployment points and search patterns

Let's assume the worst case scenario, where the main ship realizes that the submersible is lost after (one hour before) the last time the submersible sent relevant position information and parameters.

Using the submersible position model from the first model, we simulated the final position of the submersible after one hour for different parameters (current speed direction) using Monte Carlo method to obtain the following particle map (9000 particles). Then we use the "convex packet" algorithm [3], which finds a minimum convex polygon with boundaries around all the given points.

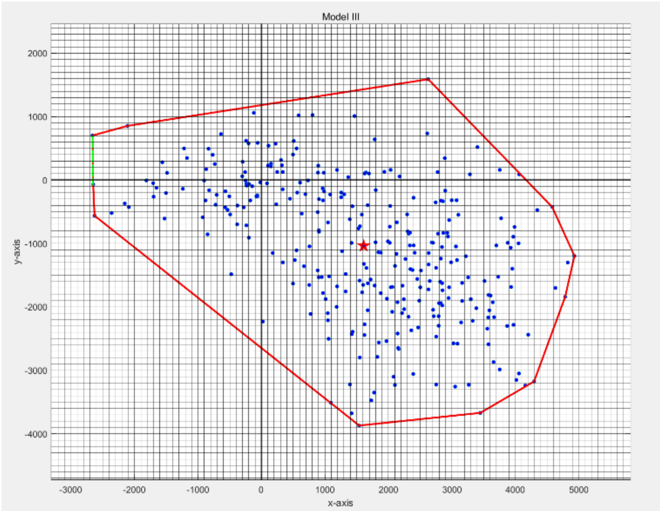


Figure 9 Submersible location scatter plot

Within this convex packet, the number of particles near the point $P_0(6550,8230)$ is the most dense, so we choose this point as the initial deployment point.

Grid search is an effective method for determining the search path of a submersible, we divide the convex packet into many small square grids whose side lengths are taken from the search range of the unmanned boat search and rescue equipment identified in 6.2, and then examine these grids one by one to find the target, it can be seen that the convex packet has been roughly divided into 125 small grids.

This method is simple and straightforward, but may require a lot of time and resources, especially when searching over a large area, so we need to optimize it.

7.2 Genetic algorithm based on solution to the traveler's problem

A genetic algorithm [4] is a heuristic search method that mimics the process of natural selection. It optimizes the search strategy by generating multiple search paths (populations) and then selecting and combining these paths based on their "fitness". Solving the traveler's problem using a genetic algorithm involves the following steps:

- **Initialization:** a set of possible search and rescue paths (called populations) is randomly generated using P_0 as a starting point.
- **Fitness function:** here the inverse of the total length of the path is used.
- **Selection:** the process of choosing well-adapted individuals from an old population to produce a new population. The selection operation of this model uses the method of fitness proportionality, which determines the probability that an individual's offspring will be left behind based on the ratio of its fitness to the sum of the fitnesses of all the individuals in that generation. Denote the number of generations as n , then the probability that an individual i is selected in the n th generation is:

$$p_{si} = \frac{f_i}{\sum f_n} \tag{30}$$

Where f_i the fitness of an individual and $\sum f_n$ is the sum of the fitness of all individuals in the generation. After determining the probability, the selection operation can be realized using the roulette wheel selection method. Python's rand( ) function is utilized to generate uniformly distributed random numbers between 0 and 1. If $p_{si} = 0.5$, then when the generated random number is between 0 and 0.5, the gene of that individual is copied, otherwise it is eliminated.

- **Crossover:** selected pathways generate new pathways (offspring) by exchanging parts of them. A crossover operation is when two paired chromosomes exchange some part of their genes with each other in a certain way, thus forming two new individuals. In genetic algorithms, n individuals in a population are randomized into $\frac{n}{2}$ pairs of paired individuals, and crossover is performed between two individuals in these pairs.

Examining two individuals in a paired group of individuals a, b, the crossover operation uses a certain way to change them into two new individuals a', b'. In genetic algorithm, the crossover operation process needs to satisfy:

$$a + b = a' + b' \tag{31}$$

Based on the above equation, the crossover operation for floating point encoding is implemented in the following way:

$$\begin{cases} a' = (1-\alpha)a + \beta b \\ b' = (1-\beta)b + \alpha a \end{cases} \tag{32}$$

Where α , β are uniformly distributed random numbers on the interval (0, r), and there is $r \le 1$, by adjusting the size of r can be adjusted by the crossover operation of the scope of change.

- **Mutation: the mutation operator is** used to adjust some of the gene values in the coding strings of an individual and to improve the local search capability of the genetic algorithm by making the individual closer to the optimal solution from a local perspective. The mutation operation changes the parameter τ of an individual, the operation to another value in the domain τ' :

$$\tau' = N(\tau, \sigma_1) \tag{33}$$

Where $N(\tau, \sigma_1)$ denotes a normally distributed random number with mean τ and variance σ_1. It can be seen that the variance operation, i.e., the change of a random perturbation centered on the current value, mainly in a small range.

- **Iteration:** for each iteration, the variant and crossover children and parents are put together and sorted, and the top w individuals with the best fitness function are selected and retained as the parents for the next iteration, completing an iterative process.

The process is repeated until the number of iterations reaches 100 and the final optimal path obtained is approximated as the shortest path solution.

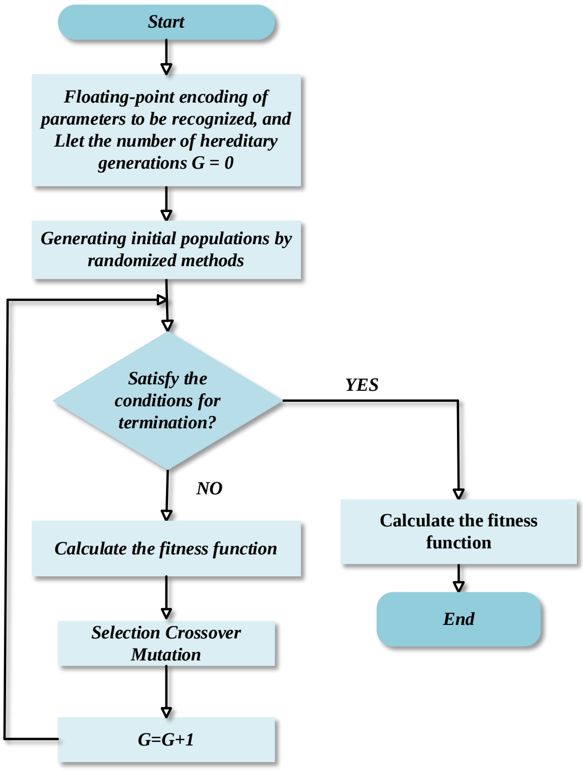


Figure 10 Genetic algorithm flowchart

However, the premise of the algorithm, focusing on the consideration of the search in the shortest time, [5] did not take into account the probability of the distribution of the submarine in the convex polygon described in 7.1, so theoretically we need to choose the path, not only to take into account the shortest total time, but also need to take into account: searching preferentially passes through the search, the probability of the submarine within the search of the area of the region with the greater probability of the submarine. Therefore, we choose the optimal 4 schemes for the 30 schemes given by the genetic algorithm: the scheme whose probability of the distribution of submarines at the point of passing through has the largest linearly decreasing trend.

The figure below shows the four scenarios , and how the probability density varies with path.

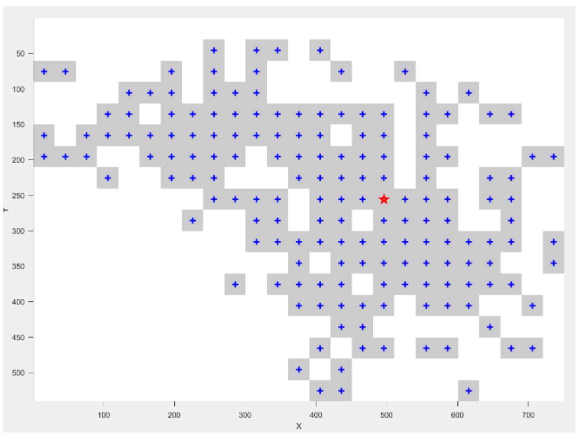


Figure 11 Predicted distribution points in unit square count plots（a）(3×3 unit square)

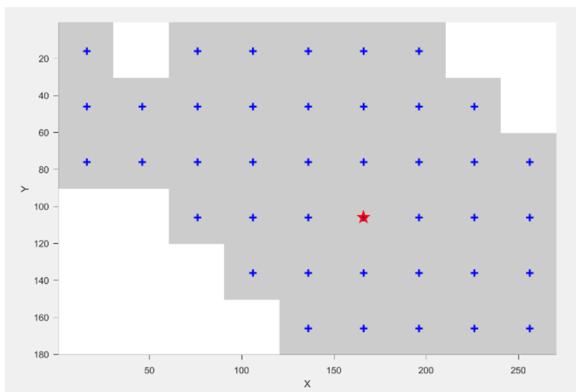


Figure 11 Predicted distribution points in unit square count plots（b）(9×9 unit square)

The number of particles in each small square is counted to estimate the a priori probability that the submersible will be found in a specific region, i.e., the probability that the submersible will land in a certain region $P(A_i) = (i = 1, 2, \cdots, n)$

$$P(A_i) = \frac{\text{Sample Size}}{\text{Overall Sample Size}} \tag{34}$$

Below are some of the results from multiple simulations of the genetic algorithm:

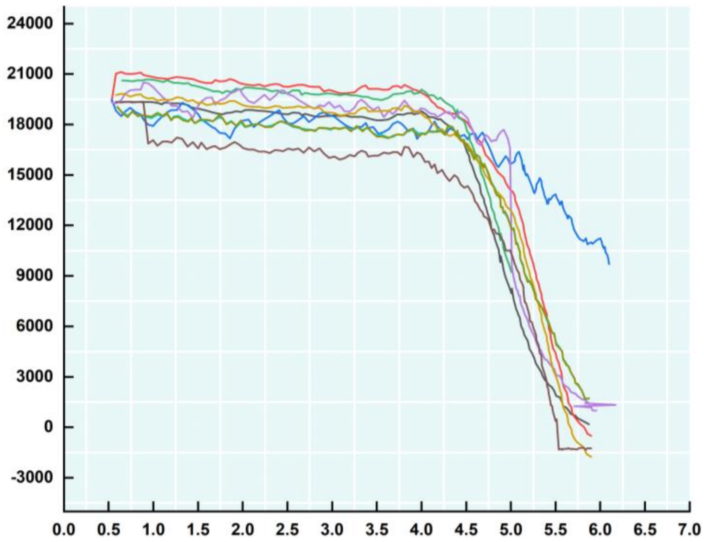


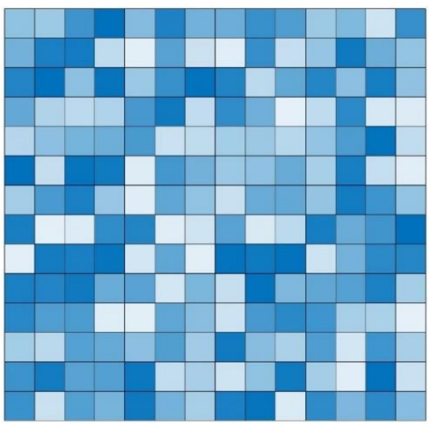
Figure 12 Partial results of the genetic algorithm

It can be seen that curve I has the most pronounced decreasing trend of probability along the position of the submersible, i.e., curve I corresponds to the best search and rescue path.

7.3 Probabilistic models based on Bayesian theory

In order to construct a probability function for finding the submersible, several key factors need to be considered, including the sweep width of the search equipment d, the speed v, and the time of the search operation t. This probability function can be built based on Bayesian theory, where the probability that the submersible will be found is constantly updated based on newly acquired information, thus increasing the search efficiency and minimizing the time required to find the submersible.

During the search and rescue process, the probability of a target being found changesis expanded, and prioritizing the area with the highest probability of discovery increases the probability of success. A Bayesian updating formula is used to update the probability of finding a submersible in a given area based on new evidence.



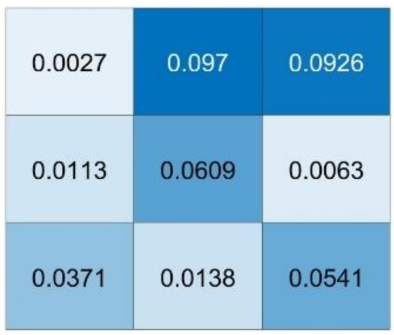


Figure 13 Probability update model（a） **Figure 13 Probability update model（b）**

$P(D|A_i)$ denotes the probability of finding a submersible in the grid A_i, which is assumed to be unchanged at $P(D|A_i)$ during each search and rescue before applying Bayes' theorem.

If no target is found in any of the searched regions, Bayes' theorem can be used to calculate the posterior probability that the target is in the region.

Calculate the probability that the target is in the searched area A_i and can be found $P(A_i \cap D)$ as:

$$P(A_i \mid D) = P(A_i) \cdot P(D \mid A_i) \tag{35}$$

Calculate the probability that the target is in the searched area A_i but not found $P(A_i \cap \bar{D})$. The sum of the probabilities that the target is in the searched region and can be found and not found is 1, therefore:

$$P(A_i \cap \bar{D}) = P(A_i) - P(A_i \cap D) \tag{36}$$

Calculate the probability that the target is in the searched area A_i but not found after all searches $P(A_i \cap \bar{D})$ [6].

$$\begin{aligned} P(A_i \mid \bar{D}) &= \frac{P(A_i \cap \bar{D})}{P(\bar{D})} \\ &= \frac{P(A_i) - P(A_i \cap D)}{\sum_{j=1}^{n} [P(A_j) - P(A_j \cap D)]} \\ &= \frac{P(A_i) - P(A_i)P(D \mid A_i)}{\sum_{j=1}^{n} [P(A_j) - P(A_j)P(D \mid A_j)]} \end{aligned} \tag{37}$$

From the above equation, it can be seen that the updating of the search probabilities can be realized by mastering only $P(A_i)$ and $P(A_i \cap D)$.

$P(D|A_i)$ Related to the performance of the search equipment, the distance traveled by the search equipment is considered under the condition that the sea sweep width ω is constant. According to the conclusions given by Koopman in Search Theory , the formula for calculating the probability of finding a target $P(D|A_i)$ is:

$$P(D \mid A_i) = 1 - e^{\frac{\omega \upsilon T}{S(A_i)}} \tag{38}$$

In the above equation, ω is the sweep width of the search device, υ is the search speed, and T is the search time.

If the submersible is not found during the first n search, it is necessary to repeat the initial probability $P(A_i)$ replacing it with the latest value. And again the search is started for the area with the highest probability of discovery, thus achieving the effect of optimizing the search area in real time.

8 Extension of Models in Other Seas

The extension of our models to other sea areas requires flexibility to adapt to the local topography of the seabed, differences in the physical properties of the sea, special climate, biodiversity, and other conditions.

The Caribbean Sea is a sea area located in the eastern part of America. Firstly, from the perspective of seafloor topography, the Caribbean Sea has an east-west trending Lepid Ridge, and there are also numerous trenches, undersea caves, and undersea volcanoes, which requires that the impact of seafloor topography and geomorphology on the detection process should be taken into account when expanding the model. In addition, the sea is always affected by the Guyana Warm Current and the North Equatorial Warm Current, which requires consideration of the influence of the special characteristics of the local ocean currents. Finally, there are many coral reefs on the seafloor, which requires consideration of the protection of the marine ecosystem.

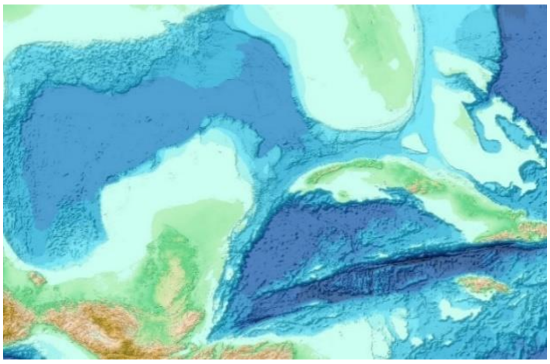


Figure 14 Topographic map of the Caribbean Sea area

Figure 15 Biological map of the Caribbean Sea area

9 Simultaneous Operation of Multiple Submersibles

9.1 Initial deployment of the submersible

The problem of multiple submersibles moving in the vicinity of a site requires setting the initial deployment distance of neighboring submersibles in terms of the safety and experience of submersible undersea tours. The following three aspects need to be considered:

- Avoidance of collisions between submersibles and the need to maintain adequate safety distances between submersibles
- The submersibles are able to communicate freely in order to share positional and environmental information in a timely manner; when a submersible encounters mechanical failure, nearby submersibles are able to go to search and rescue in a timely manner.
- Striving to maximize the area covered by the expedition reduces repetitive exploration between submersibles and maximizes the visitor's undersea sightseeing experience.

9.2 Mechanisms for matching submersibles in emergencies

To simplify the problem for the sake of simplicity, we study a matched group of four submersibles at a given moment in time. We believe this number is sufficient to demonstrate the complexity of multiple submersibles working together without complicating the problem.

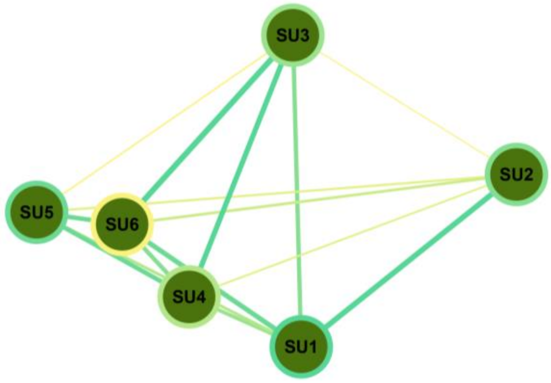


Figure 16 Matching unit operating mode

In response to the problem of multiple submersibles in the vicinity of the same site, we have established the following mechanism: The main ship finds that there is a possibility of loss of connection or mechanical failure of a certain submersible, and immediately notifies other submersibles to jointly carry out the rescue according to the unit matched before the loss of connection. Any one of the submersibles is matched to the three surrounding submersibles B,C,D according to their relative positions to form a group of four submersibles. Matching principle: The three submersibles are the nearest three, and their relative positions satisfy the overall "wraparound situation" of B, C, and D to A. In addition, the matching unit will be updated as the relative positions change over time. In addition, as the relative positions of the submersibles change over time, the matching unit will be updated.

9.3 Innovative ideas for multi-submersible search and rescue

On the basis of model three, multiple submersibles launch search and rescue on the same target at different locations at the same time, and strive to shorten the search and rescue time as much as possible on the premise of maximizing the success rate of search and rescue.

Combined with the military field of the "package" military thinking, here for a simple migration, B, C, D, three A maximum probability distribution point position as the center of the ball, search, three search track with the superposition of time can maximize the coverage of the entire track corresponding to the sphere of the sphere; at the same time, let the radius of the track is constantly shrinking, the overall formation of the encirclement of A, that is, the so-called "package" method. That is the so-called "encircling method".

Figure 17 Matching unit joint search and rescue model diagram

10 Model Testing

10.1 Sensitivity analysis

In Section 5.2.1, we parametrically estimated the drag coefficient of the submersible C_d to be 0.42, but the actual drag coefficient of a submersible depends on its shape, surface properties, and the nature of the fluid, and can only be obtained by means of measurements, which are generally considered to be in the range of 0.3-0.5 for submersibles assumed to be cylindrical.

From $F = \frac{1}{2} C_d \rho A \nabla V^2$, the force of the current on the submersible shows a linear relationship with the drag coefficient, so a change in the drag coefficient theoretically changes the current's power on the submersible significantly.

We need to test the results of model one by traversing C_d from 0.3 to 0.5 values in steps of 0.02 to make the longitudinal displacement D_y and transverse displacement D_x of the submersible in relation to the drag coefficients after one hour, respectively.

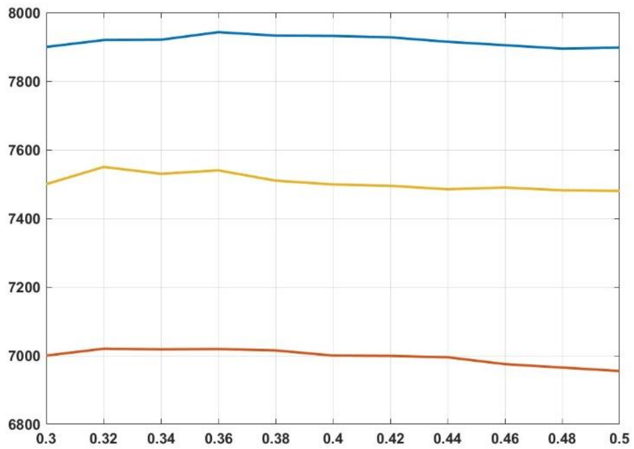


Figure 18 C_d sensitivity test

The study shows that the trends of the longitudinal and transverse displacements of the submersible are relatively smooth, and also proves the robustness of Model I.

10.2 Robustness analysis

In Model 3, the magnitude of the velocity of the ocean currents is important in determining the search range of the SAR vessel. In general, the greater the velocity of the current, the greater the range of motion of the submersible, thus expanding the search and rescue range.

Therefore, we modeled the migration of the submersible by considering the current velocities under different initial distributions, denoting the final distribution of the first i submersible as $\Phi_i\left(x, y \mid x, y \in \Omega\right)$,Ω denoting the set of all possible current velocities, and then calculating the goodness-of-fit between any two distributions $RMSE_{ij}$:

$$\mathrm{RMSE}_{ij}=\sqrt{\frac{1}{N}\sum_{(x,y)\in\Omega}[\Phi_i(x,y)-\Phi_j(x,y)]^2} \tag{39}$$

Form $RMSE_{ij}$ into a paired sequence to make a normal distribution test Q-Q plot:

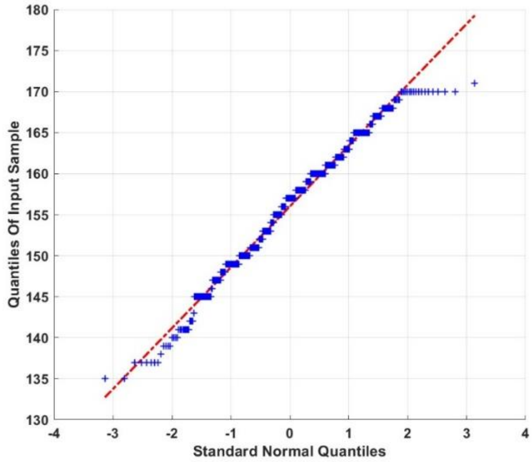


Figure 19 Q—Q plot of RMSE v.s standard normal

It can be seen that the scatter points of the plots are distributed in a straight line, indicating that the overall fitting is stable. It means that under the premise of different initial current velocity distributions, similar distributions of the possible positions of the final submersible can be obtained, which verifies the decisive role of the currents on the position of the submersible and reflects the stability of the model.

11 Strengths and Weaknesses

11.1 Strengths

- Task1: Combining the physical theories of kinematics and hydrodynamics, comprehensively considering the influence of sea currents and seawater density environmental factors on the prediction of the trajectory of the power operation, and using the algorithm combining the Bootstrap method and the Monte Carlo algorithm, which is able to simulate the complex undersea current speed situation well, and ultimately is able to realize the accurate prediction of the position of the faulty submersible.
- Task2: Using the TOPSIS algorithm combined with the entropy weight method, we are able to synthesize the advantages and disadvantages attributes of the 72 options, and are able to make a comprehensive, objective, and rational option selection.
- Task3: On the basis of the predictions of model 1, using the convex packet algorithm and the genetic algorithm, we are able to find the optimal solution path to find the problematic diver.
- Task4: Multi-submersible problem is based on the model three considerations and innovatively proposes the modeling idea of realizing rescue in the dimension of four-dimensional space.

11.2 Weakness

Detailed data on currents, temperatures, and topography of the Ionian Sea would make the modeling process more accurate in analyzing the actual situation.

Our model does not take into account too much the economic efficiency of the company, but mainly the safety dress code from the point of view of the development of safety procedures in the event of machine failures in submersibles.

12 Appendices

[1] Yuan Guang Research on predictive control method of manned submersible navigation attitude model [D]. Northeastern University,2019.

[2] LI Dejun,ZHANG Wei,WANG Lei et al. Simulation study of submersible motion under the action of sea current[J]. Ship Science and Technology,2023,45(11):13-16.

[3] LI Dejun, ZHAO Qiaosheng, HE Chunrong et al. Motion simulation and experimental verification of deep-sea manned submersible[J]. Ship Science and Technology,2021,43(11):62-65.

[4] XU Yufei,PAN Changpeng,WANG Lei. Selection and optimization of search and rescue area of shipboard helicopter[J]. Ship Electronic Engineering,2016,36(12):26-28+126.

[5] XU Bochang, WANG Shufu, TIAN Suzhen. Simulation of vertical distribution of seawater density (σ_t) in shallow sea[J]. Huang Bohai Sea Ocean,1983(01):9-12.

[6] WANG Xiuling,YIN Yong,ZHAO Yanjie et al. A review of unmanned boat maritime search and rescue path planning technology[J]. Ship Engineering,2023,45(04):50-57.DOI:10.13788/j.cnki.cbgc.2023.04.09.

Memo

To: Government Regulator

From: Government Regulator

Date: Jan 5^{th}, 2024

Subject: MCMS Corporation's Program on the Preparation of Safety Procedures for Underwater Tourism Operations by Manned Submersibles

Dear Government Regulator,

The Ionian Sea, which is the easternmost part of the Mediterranean Sea, is considered one of the most beautiful seas in the Mediterranean and is rich in marine tourism. Our company, Maritime Cruises Mini-Submarines Corporation (MCMS), is a Greek company dedicated to constructing manned submersibles. With the geographical advantage of having the Ionian Sea to the west, the Company proposes to open a tour of the undersea hull wrecks of manned submersibles in the Ionian Sea. The following is a relevant elaboration of the scenario in which the manned sightseeing submersible designed by our company encounters a malfunction and enters into a safety procedure.

The sightseeing manned submersible is first relocated from the main ship to the sea near the sightseeing point and then released, at this time, the submersible is detached from the main ship for undersea sightseeing. Considering the possible loss of power or loss of connection encountered by the submersible during the sightseeing process, our company builds three systems based on three models, namely, Preparation System A, Prediction System B, and Search and Rescue System C, and then proposes the following safety procedures.

Safety Procedures Overview

- Preparatory System A

Preparatory system A provides the equipping program for the main ship to carry additional search equipment and the preparatory program for the rescue ship to carry search and rescue equipment. This achieves efficient sea rescue and maximizes the rescue success rate.

- Prediction System B

①Contact mechanism: the submersible in the dive, sightseeing process, a certain period to send signals to the main ship, real-time to maintain efficient contact. At this time, the submersible will send data information to the main ship, including latitude and longitude position, depth, seawater temperature, seawater flow rate, and various parameter indicators of the submersible itself, to ensure that the main ship accurately assesses the underwater safety condition of the operating submersible.

②Analyzing mechanism: The analyzing system combines kinematics and hydrodynamics, comprehensively considers the influence of the current environment on the trajectory of the submersible, and can predict the maximum range where the faulty submersible is located in a certain time. After evaluation, our successful prediction rate reaches more than 90%, and we can know its position very accurately. By analyzing the data fed back from the submersible, the main ship can predict the change of the submersible's position over time, to be able to carry out search and rescue at sea quickly after the emergence of the external.

- Search and Rescue System C

①Single-submersible operation search and rescue mechanism: when there is only one submersible operation and failure, the main ship determines the position of the faulty submersible, the main ship will carry out search and rescue following the specified search mode: calculate the position where the faulty submersible has the greatest probability to appear and start from there, and according to the optimal path measured by the system, go through the positions where the submersible has the greatest probability to appear in order so that the faulty submersible can be searched and rescued in the shortest possible time. The system can search and rescue the faulty submersible in the shortest time.

② Multi-submersible operation search and rescue mechanism: when multiple submersibles are operating at the same time and one of them fails, the main ship will notify the closest submersible to the faulty submersible before the loss of the connection, i.e., the rest of the submersibles that are operating normally in the same matching unit to launch a joint rescue, and carry out the search and rescue according to the search and rescue mode of "packet clamping".

Looking forward to the opportunity to further discuss our proposal and work together towards enriching Greece's marine tourism landscape.

Sincerely,

Michael Jordan

Michael Jordan
123 Main Street, Athens, Greece
Maritime Cruises Mini-Submarines Corporation
Contact information:+30 210 1234567

Report On Use Of AI

OpenAI ChatGPT (Nov5, 2023 version, ChatGPT-4, ChatGPT-3.5)
Query1: Sea floor landforms of the Caribbean Sea
Output1: The Caribbean Sea is known for its diverse and fascinating sea floor landforms. Here are some notable ones: 1. Trenches: The Caribbean Sea features several trenches, which are long, narrow depressions in the sea floor. The deepest trench in the Caribbean is the Cayman Trench, located near the Cayman Islands. It reaches depths of over 7,686 meters (25,217 feet) and is one of the deepest points in the world's oceans. 2. Seamounts: Seamounts are underwater mountains that rise from the sea floor. The Caribbean Sea has numerous seamounts, including the Saba Bank, the largest submerged atoll in the Atlantic Ocean. Seamounts provide important habitats for marine life and can attract a variety of species. 3. Coral Reefs: The Caribbean is renowned for its extensive coral reef systems. These reefs are formed by the accumulation of coral skeletons over thousands of years. The Mesoamerican Barrier Reef System, also known as the Great Mayan Reef, is the largest coral reef system in the Caribbean and the second-largest in the world. It stretches over 1,000 kilometers (620 miles) along the coasts of Mexico, Belize, Guatemala, and Honduras. 4. Abyssal Plains: Abyssal plains are flat, sediment-covered areas of the deep sea floor. While the Caribbean Sea is not primarily characterized by abyssal plains, some can be found in certain areas. These plains are typically located at great depths and are important for sediment deposition and the cycling of nutrients in the marine ecosystem. 5. Canyons: Submarine canyons are deep, steep-sided valleys that cut through the continental shelf and slope. The Caribbean Sea has several canyons, including the Puerto Rico Trench and the Mona Canyon. These canyons often serve as pathways for nutrient-rich water and can support diverse marine life. 6. Volcanic Features: The Caribbean Sea is part of the Caribbean Plate, which is surrounded by volcanic activity. Volcanic features such as underwater volcanoes, volcanic arcs, and volcanic islands can be found in the region. For example, the Lesser Antilles arc consists of a chain of volcanic islands and submerged volcanoes. These sea floor landforms contribute to the unique and vibrant marine ecosystem of the Caribbean Sea, supporting a rich diversity of marine species and providing important habitats for various organisms.
Query2: Template format of the memo
Output2: [Your Company's Logo] [Date] MEMO To: [Recipient's Name] From: [Your Name] Subject: [Subject of the Memo]

[Body of the Memo]
[Closing]
[Your Name]
[Your Position]
[Your Contact Information]

Query3:

```
A = load('BB.txt');
B = zeros(6, 9);%18 25
for i = 1:6
for j = 1:9
sum_9x9 = sum(sum(A((i-1)*9+1:i*9, (j-1)*9+1:j*9)));
B(i, j) = sum_9x9;
end
end
grid_matrix = flipud(B);
grid_size = 30;
image_width = size(grid_matrix, 2) * grid_size;
image_height =size(grid_matrix, 1) * grid_size;
image = ones(image_height, image_width);
for i = 1:size(grid_matrix, 1)
for j = 1:size(grid_matrix, 2)
if grid_matrix(i, j) ~= 0
x = (j - 1) * grid_size + 1;
y = (i - 1) * grid_size + 1;
image(y:y+grid_size-1, x:x+grid_size-1) = 0.8;
end
end
end
figure;
imshow(image);
axis off;
shadow_centers = [];
for i = 1:size(grid_matrix, 1)
for j = 1:size(grid_matrix, 2)
if grid_matrix(i, j) ~= 0
x = (j - 1) * grid_size + 1;
y = (i - 1) * grid_size + 1;
image(y:y+grid_size-1, x:x+grid_size-1) = 0.8;
center_x = x + grid_size/2;
center_y = y + grid_size/2;
shadow_centers = [shadow_centers; [center_x, center_y]];
end
end
```

```
end
figure;
imshow(image);
axis off;
hold on;
plot(shadow_centers(:,1), shadow_centers(:,2),...
'b+','MarkerSize',8,'LineWidth',2);
ax=gca;
ax.XAxis.Visible='on';
ax.YAxis.Visible='on';
xlabel('X');ylabel('Y');
plot(166,106, 'rp', 'MarkerSize', 10, 'LineWidth', 2)
shadow_centers
```

I need to combine the data from the grid_matrix corresponding to the middle coordinates of the shaded squares into a matrix

You can use the following code to merge the grid_matrix data corresponding to the coordinates in the middle of the shaded square into a matrix.

Output3:

```
shadow_data = [];
for i = 1:size(shadow_centers, 1)
center_x = shadow_centers(i, 1);
center_y = shadow_centers(i, 2);
grid_row = ceil(center_y / grid_size);
grid_col = ceil(center_x / grid_sizdata = grid_matrix(grid_row, grid_col);
shadow_data = [shadow_data; data];
end
```

You can use the following code to merge the grid_matrix data corresponding to the middle position coordinates of the shaded square into a matrix: Thus, you will have a matrix named 'shadow_data' that contains the grid_matrix data corresponding to the middle position coordinates of the shaded square.

Query4: Can I just consider the motion of a submersible in seawater with 3 degrees of freedom?

Output4:

Yes, you can choose to consider the motion of the submersible in seawater with only 3 degrees of freedom (DoF). In robotics and dynamics, Degree of Freedom (DoF) refers to the number of ways to move independently in a physical system. For an underwater vehicle such as a submersible, its motion in three-dimensional space can typically have six degrees of freedom: three translational degrees of freedom and three rotational degrees of freedom.

The three translational degrees of freedom include:

Longitudinal movement (Surge): movement in the forward direction of the submersible.

Sway: Horizontal movement perpendicular to the forward direction of the submersible.

Vertical movement (Heave): movement in ascending and descending.

The three rotational degrees of freedom include:

Roll: rotation around the axis in the forward direction of the submersible. Pitch: rotation around the horizontal axis of the submersible, affecting ascent and descent. Yaw: rotation around the vertical axis of the submersible, changing the forward direction. In some cases, in order to simplify the analysis, you can consider only the three translational degrees of freedom of the submersible (longitudinal, lateral and vertical movement) and ignore the rotational degrees of freedom. This simplification usually applies in the following cases: The study focuses on changes in the position of the submersible, rather than changes in its attitude (orientation). The effects of rotational motion are small or can be ignored during the preliminary analysis phase. Preliminary evaluation of the engineering design or control system, where the primary focus is on the basic navigation and positioning of the submersible. However, it is important to note that while only three translational degrees of freedom may be considered in some studies or applications, a complete dynamics model of a submersible and the design of the control system typically requires consideration of all six degrees of freedom to ensure accuracy and operational flexibility. This comprehensive consideration enables more accurate modelling and prediction of the behaviour and performance of submersibles in complex marine environments.
Query5:International Standard Units of Pressure, Temperature and Salinity
Output5: The international standard units of pressure, temperature and salinity are as follows: Pressure: In the International System of Units (SI), the unit of pressure is the Pascal (symbol Pa), with 1 Pa equal to 1 Newton per square metre (N/m^2). In oceanography, depth is often used as an indirect indication of pressure, so the bar is also commonly used as a unit of pressure, 1 bar = 100,000 Pa = 100 kPa. Temperature: In the International System of Units, the unit of temperature is the Kelvin (symbol K). In everyday life and scientific research, temperature is often expressed in Celsius (symbol °C). Salinity: Salinity is a physical quantity that indicates the salt content of water. In the past, salinity was often expressed as "parts per thousand" (symbol ‰), but this method relied on the measurement of conductivity. In modern oceanography, the international standard unit of salinity is Absolute Salinity (symbol: g/kg), which is a mass-based measure of how many grams of salt are present per kilogram of seawater. These units are widely used internationally and are the standard in scientific research and technical applications. The use of these international standard units is very important when carrying out research in the fields of oceanography, meteorology, environmental science, etc., as they guarantee the accuracy of data and consistency of comparisons.
Query6: What is the difference between the equipment required for a rescue boat and that required for a main boat?
Output6:

Rescue vessels and primary vessels (or mother ships) are fundamentally different in the tasks and functions they perform, which leads to significant differences in the equipment and systems they require. Below are some of the basic differences:

Equipment Required for Rescue Vessels

Rescue vessels specialize in search and rescue, providing emergency assistance, and conducting maritime safety operations, so the equipment they are equipped with typically includes:

Search and Rescue Equipment: This includes radar, sonar systems, search and rescue helicopters (if deck space is available), rope throwers, life rafts, life buoys, etc. to facilitate rapid location and rescue of persons in distress.

Communication equipment: advanced communication systems including satellite telephones, short and long wave radios, emergency locator beacons (EPIRBs) and personal locator beacons (PLBs), etc. to ensure communication with the outside world under all conditions.

Medical equipment: includes first aid equipment, medicines, medical instruments and possible medical evacuation facilities to provide emergency medical assistance.

Emergency assistance equipment: e.g. fire-fighting equipment, underwater breathing apparatus, cutting and welding tools, etc. to deal with emergencies.

Rescuer equipment: specialized diving equipment, rescue suits, harnesses, etc. to protect rescuers.

Equipment required for the main ship

As a support and operation center, the equipment of the main ship or mother ship pays more attention to the ship's daily operation and ocean-going capability:

Navigation equipment: GPS navigation system, radar, sonar, automatic identification system (AIS), etc. to ensure navigation safety.

Communication system: including satellite communication, radio communication equipment, etc. to ensure communication with other vessels and shore-based facilities.

Power and propulsion systems: powerful engines, standby engines, propulsion systems, etc., to ensure that the vessel can navigate in a variety of ocean conditions.

Life support systems: including food and potable water storage, accommodation facilities, waste disposal systems, etc., to ensure the daily life needs of the crew.

Cargo and equipment carrying systems: booms, cargo holds, etc. for transporting and carrying submersibles, dinghies, research equipment or other special cargo.

In summary, the equipment of a rescue vessel is more focused on search and rescue and the provision of emergency assistance, whereas the equipment of a main vessel is focused on ensuring navigational safety, communications and day-to-day operations. Different types of vessels will have different equipment configuration needs based on their specific missions and roles.